

Yash Jain

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EDUCATION

University of Waterloo

Candidate for Bachelor of Software Engineering, CGPA: 3.9/4.0

- USA Computing Olympiad Gold, Engineering Ambassador

Waterloo, ON

Sep 2025 – Apr 2030

EXPERIENCE

Software Research Engineer

Space and Atmospheric Instrumentation Lab

Jun 2023 – Aug 2024

Daytona, FL - Remote

- Engineered a real-time telemetry parsing engine for NASA funded SEED mission rockets, leveraging Python and NumPy to process asynchronous bit-masked packets at a throughput of 620 KB/s
- Reduced missed packets by 90% by refactoring the legacy MATLAB plotting suite to VisPy and vectorizing raw sensor data processing using NumPy, achieving higher accuracy and a 4x increase in plot frame rates
- Generalized the program to multiple missions through decoupling data definitions from the codebase with Excel, removing the need for researchers to code and reducing human error during missions

Software Developer

Waterloo Rocketry

Sep 2025 – Present

Waterloo, ON

- Developed a real-time network bridge to transmit live rocket telemetry to ground control dashboards, integrating a type-safe serialization protocol using Pydantic to validate 30+ sensor fields and eliminate malformed packets.
- Created a pre-competition confidence check by creating a Pytest testing framework, achieving high code coverage for message parsing, which was also used to identify three deprecated protocols in the codebase

Team Captain and Programmer

Salish Robotics Team

Sep 2022 – Jun 2025

Vancouver, BC

- Trained autonomous navigation routines in C++ by implementing PID controllers for precision steering and Pure Pursuit for smooth path-tracking, optimizing for consistency during competition
- Managed the 3D design lifecycle in OnShape to coordinate a team of 8 by defining clear mechanical interfaces that allowed hardware and software development to proceed in parallel
- Secured 2 league awards by collaboratively analyzing past failures and adaptively strategizing (judges and create award)

PROJECTS

Wunder Challenge | *PyTorch, ONNX, Pandas*

- Designed a hybrid CNN-LSTM architecture in PyTorch to extract features from high-frequency Limit Order Book data, capturing both local price volatility and long-term temporal dependencies
- Optimized signal-to-noise ratios by implementing custom preprocessing for data snapshots, outperforming 96% of 3,000+ competitors in a Pearson Correlation-based mid-price prediction challenge

A-List Housing | *Python, TypeScript, React, Flask, Selenium, Auth0*

- Engineered a full-stack real estate affordability platform utilizing Flask, TypeScript, and Auth0
- Created an engine to forecast macro-economic housing trajectories by implementing an XGBoost hedonic pricing classifier trained on 100k+ property listings
- Developed a k-nearest neighbours recommendation feature to identify comparable property sales, utilizing historical market data gathered through Selenium web scraping

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript, Java, MATLAB, SQL, Bash

Technologies: NumPy, PyTorch, Pandas, Websockets, React, Node.js, Flask, PyQt, VisPy, Matplotlib, PyTest